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ARTICLE

Heritability and selection of strategic traits in the NFL coaching tree

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Abstract

Expertise passes from one practitioner to the next, but the lineages it travels along are rarely observable. Professional sport is an exception: the record of who coached under whom is public, and the behavior is logged play by play. Cultural-evolution research has shown how winning sport tactics spread sideways among contemporaries; we ask instead what descends the mentor–protege line. Treating the National Football League coaching tree as a cultural pedigree, we take ten preregistered standardized play-calling traits and estimate, for each, its transmissibility (h^2, mentor-to-protege resemblance from a Bayesian errors-in-variables parent-offspring model) and its selection (S, its association with coach wins above replacement, WAR, a coach's contribution net of roster and schedule). Each trait is repeatable (repeatabilities 0.34 to 0.58), yet transmissibility and selection come apart. On offense, schematic-identity traits transmit strongly but are at most weakly selected: shotgun $h^2 = 0.45$ (95% HDI $[0.27, 0.65]$) with $S = 0.10$ ($[0.00, 0.20]$), and tempo with no detectable selection ($S = 0.02$). Offensive aggression is the mirror image, repeatable (0.40) and selected ($S = 0.18$) yet not transmitted ($h^2 = -0.15$). Defensive aggression, which bundles box-stacking, pass rush, and man coverage, most clearly both transmits ($h^2 = 0.49$) and wins ($S = 0.17$), consistent with its mixing schematic identity and in-game approach. The predicted negative heritability–fitness relationship did not appear (Spearman +0.31, $[-0.52, 0.72]$); with few transitions this is inconclusive, and transmission and selection appear largely decoupled rather than traded off. Coaching lineages transmit a mentor's system more reliably than his success: a protege inherits how his teams play, while the offensive aggression that wins spreads sideways among contemporaries rather than down the tree.

Keywords: cultural evolution, social learning, vertical transmission, heritability, cultural fitness, sport

1. Introduction

Expertise passes from one practitioner to the next, and where practitioners learn from identifiable teachers, that transmission runs along a traceable lineage. Coaching in the National Football League (NFL) is a case where the lineage is unusually legible. An NFL staff is organized as a strict hierarchy that is consistent across teams and eras: a head coach leads the offensive, defensive, and special-teams coordinators, who in turn lead the position coaches under them, and coaches climb that ladder and move between teams in a record documented season by season. The result is a "coaching tree": a directed network of mentor–protege ties along which styles of play, and coaching ability itself, are widely believed to descend. Popular accounts speak of the Walsh tree or the Belichick tree much as a breeder speaks of a bloodline, and teams appear to hire on that pedigree logic, in the belief that working under a successful coach transfers winning approaches and strategic expertise.

Cultural evolution treats socially learned behavior as a second inheritance system, one that, like the genetic system, is transmitted, varies, and is subject to selection (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981; Henrich, 2016). Much of what distinguishes an expert is acquired by learning from more accomplished practitioners rather than reinvented from scratch, so specialized skill tends to descend along chains of social transmission (Henrich, 2016). Measuring that vertical

transmission is nonetheless hard: it requires observing the same behavior in both teacher and learner, on a comparable scale, with the lineage known, and such behavioral lineages are seldom legible in humans. Professional sport offers an unusually clear window onto cultural transmission for exactly this reason: the record of who coached under whom is public, and the behavior itself is logged play by play.

Recent work has begun to treat sport tactics as a cultural-evolutionary system. Mesoudi (2020), in this journal, analyzed the cultural evolution of association-football (soccer) formations across more than nine thousand matches in five European leagues and showed that a manager's use of the era's dominant formation (the 4-2-3-1) tracked both his own recent use of and success with it and how widely it was used across contemporary managers. Formation choice was guided by strategic social learning, and a successful tactic diffused as managers copied it from their peers: a horizontal channel, transmission among contemporaries. That study measured the horizontal channel and named two things it could not reach as open questions: transmission down mentor–protege lineages (i.e., the vertical channel) and the flow of influence along the social-network ties that connect managers, for which the author offered the lineage of managers who carry Marcelo Bielsa's approach as the motivating example. The present paper takes up those questions in the NFL. Cultural transmission theory distinguishes vertical transmission, from a parent-like model to a learner, from horizontal transmission among contemporaries, and the two need not carry the same traits (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981). As we will show, the vertical and horizontal channels in coaching move different strategic traits, so the two studies are complementary halves of one picture.

To ask how faithfully behavior descends such a lineage, we borrow the toolkit quantitative genetics built for that decomposition, which rests on two quantities. The first, repeatability, is the share of a trait's variance that is a stable property of the individual rather than occasion-to-occasion noise, and it sets an upper bound on how much of the trait can be transmitted at all. The second, heritability, is the share actually passed to descendants; in a parent-offspring design it is read directly from the resemblance between parent and offspring phenotypes, with no need for an assumed relatedness matrix (Falconer & Mackay, 1996). The same variance-partitioning logic extends from classical pedigrees to repeated measures through the mixed-model animal model (Nakagawa & Schielzeth, 2010; Wilson et al., 2010), and dual-inheritance theory has long held that it applies to culturally transmitted traits as well as genetic ones (Cavalli-Sforza & Feldman, 1981). The coaching tree suits this design well, because both the lineage and the phenotype are observed directly rather than inferred.

We use these tools to place each trait on two axes. The first is transmissibility, the cultural analog of heritability h^2: how strongly a protege's own head-coaching style resembles the style of the coach they apprenticed under. The second is selection S: how strongly a trait is associated with a coach's success, measured by wins above replacement (WAR). WAR estimates a coach's own contribution to winning rather than his roster's or schedule's; we take up why it is the appropriate fitness proxy, and its limits, in the Methods. In biological populations the two are famously in tension. Fisher's fundamental theorem implies that selection consumes the additive genetic variance of the traits most closely tied to fitness, so those traits should be the *least* heritable (Fisher, 1930). The prediction holds across taxa: Mousseau and Roff (1987) found life-history traits, those nearest fitness, markedly less heritable than morphological ones, and Kruuk et al. (2000) report a rank correlation near -0.8 between the heritability of a suite of red-deer traits and their association with fitness. Whether an analogous tension arises in a cultural system, where inheritance is social learning rather than meiosis and the variance-depleting logic of Fisher's theorem need not apply, is unknown.

We proceed in three steps. First, we ask whether these play-calling tendencies are genuine, stable properties of a coach at all rather than artifacts of a roster or a season. Second, we ask whether they transmit from mentor to protege down the lineage (the heritability h^2). Third, we ask whether they predict a coach's success (the selection S). The question that ties the three together is whether the traits

that transmit are the same traits that win. We separate strategy into two families. *Approach-to-scenarios* traits describe how aggressively a coach attacks game situations (going for it on fourth down, passing in neutral situations, throwing beyond the first-down marker, attempting two-point conversions, and on defense loading the box, sending extra rushers, and playing man coverage). *Schematic-identity* traits describe the formational signature of a team (how much it operates from shotgun, and how fast it plays). The split is cleanest on offense; as we discuss below, defensive aggression mixes an attitudinal and a schematic component. Our preregistered hypotheses were:

H1. Approach traits are more strongly selected (predict WAR), while identity traits are more strongly inherited (transmit down the tree).
H2. Offensive aggression is a repeatable individual trait that is nonetheless not heritable: its within-coach repeatability clearly exceeds its transmissibility.
H3. Across the ten sub-traits, heritability and selection are negatively related: the traits that transmit most strongly are not the ones that predict winning.

These predictions were registered before any confirmatory quantity was computed on the final measurement pipeline (Section 2).

2. Methods

2.1 Coaching phenotypes ("genes")

We construct standardized coach-season phenotypes from play-by-play data for the 2006–2024 NFL seasons (obtained through `nfl_data_py`; more than 600,000 plays). Each of the ten traits (Table 1) is the residual of a play-level expectation model: a gradient-boosted tree ensemble trained on pre-play game context only (down, distance, field position, score, clock, and environment), deliberately excluding season, week, and coach identity, so the baseline reflects what an average coach would do in identical circumstances and is time-invariant. The models are classifiers for the discrete decisions and regressors for the continuous ones (pace and the box and rusher counts); situational context explains most of the variance in each decision, leaving the smaller, coach-specific residual we read as tendency. Writing a_{cj} for coach c's action on play j of that coach's n_c plays in a season (an indicator for the discrete decisions, a continuous value for pace and the box and rusher counts) and $\hat{a}_{cj}$ for the model's expectation given that play's pre-snap context, the raw gene is

$$g_c = \frac{1}{n_c}\sum_{j=1}^{n_c} a_{cj} - \frac{1}{n_c}\sum_{j=1}^{n_c} \hat{a}_{cj}, \tag{1}$$

and the phenotype is its standardization across coach-seasons, $\tilde{g}_c = (g_c - \bar{g})/\hat{\sigma}_g$, where $\bar{g}$ and $\hat{\sigma}_g$ are the sample mean and standard deviation of g over all coach-seasons; positive values denote more aggressive, faster, or more shotgun-oriented play than context predicts. This residual construction isolates a strategic tendency from forced circumstance: a coach who goes for it often only because he is always trailing is not thereby aggressive.

The ten sub-traits comprise four offensive-aggression decisions (going for it on fourth down, passing versus running in neutral situations, throwing a pass beyond versus short of the first-down marker, and attempting a two-point conversion), two tempo behaviors (no-huddle use and seconds between snaps), three defensive behaviors (defenders in the box, number of pass rushers, and man versus zone coverage), and shotgun formation use; each model's type and out-of-sample performance are listed in Table 1. The pass-rush regressor is a partial exception, with a low R^2 (about 0.05): pre-play context predicts little of the play-to-play variation in how many defenders a team sends. Because the gene is a mean deviation rather than an explained-variance quantity, a low R^2 does not undermine it: the model still supplies an unbiased expectation to residualize against, and the residual retains a repeatable, transmissible coach signal (Table 3). The expectation models, the leakage

Table 1. The ten frozen sub-traits: gene-level composite, the play-level expectation model whose residual defines the phenotype and its out-of-sample performance (AUC for classifiers, R^2 for regressors, under grouped resampling that holds out whole team-years), and the data window. All transitions tested are offensive-coordinator-to-head-coach except the three defensive traits, which are defensive-coordinator-to-head-coach.

Sub-trait	Composite	Model	Performance	Window
Fourth-down	Offensive aggression	classifier	AUC 0.97	2006–2024
Pass-heavy	Offensive aggression	classifier	AUC 0.78	2006–2024
Deep-pass	Offensive aggression	classifier	AUC 0.73	2006–2024
Two-point	Offensive aggression	classifier	AUC 0.93	2006–2024
No-huddle	Tempo	classifier	AUC 0.77	2006–2024
Pace	Tempo	regressor	R^2 0.43	2006–2024
Box-stacking	Defensive aggression	regressor	R^2 0.42	2016–2024
Pass-rush	Defensive aggression	regressor	R^2 0.05	2016–2024
Man-coverage	Defensive aggression	classifier	AUC 0.68	2018–2024
Shotgun	Shotgun	classifier	AUC 0.81	2006–2024

controls that keep a coach's own tendencies out of the yardstick that measures them, and the reliability weighting are detailed in the next two subsections.

Every play is attributed to the head coach who led the team for that game, so a within-season head-coaching change splits the season between the two coaches at the game of the change. The expectation models are then fit with the scoring coach held out and with cross-fitting folds balanced across eras, so each play is scored by a model that never trained on that coach.

Ten sub-traits are frozen for the primary analysis (Table 1), grouped into four gene-level composites (offensive aggression, tempo, defensive aggression, and shotgun) that serve as labeled exemplars but are not counted as independent points in the cross-trait test.

2.2 Expectation models, leakage control, and feature selection

Each of the ten expectation models is a gradient-boosted tree ensemble, consistently the strongest model class for tabular, point-in-time prediction of this kind (Chen & Guestrin, 2016). Three safeguards make the residuals trustworthy as coach phenotypes.

First, leakage-free scoring. Because a gene is a residual against a model's prediction, it is meaningful only if a coach-season is scored in exactly the feature space the model was trained in. We fit every preprocessing step, the categorical encoders, the median imputer, and a truncated-SVD reduction, on the training split alone, persist them, and apply them in transform-only mode when scoring the full play-by-play record; this removes a train/serve mismatch that would otherwise place genes in a different space than the one the models learned. Imputation is a minor step here: apart from temperature and wind, which are absent for indoor games and carry little weight in these models, the retained fields are all but complete, so median filling touches only a small share of entries and leaves the genes essentially unchanged. Coach identity, season, and week are excluded as predictors, so no model can recover a coach from a name or the calendar, and every split is made at the team-year group level so that no team or coach appears in both training and evaluation.

Second, coach-held-out cross-fitting. A model trained on a coach's own plays would partly fit that coach's tendencies and shrink the very residual we want to measure. The predictions used to compute a gene are therefore out-of-fold, from a group k-fold that leaves out whole coaches (the head coach of the offense for the offensive genes, of the defending team for the defensive genes), reusing the tuned hyperparameters, with folds balanced across eras (stratified by career midpoint). Each coach-season is scored by a model that never saw that coach, so the residual reflects deviation from a coach-blind baseline. The performance figures in Table 1 are out-of-fold estimates under

grouped resampling that holds out whole team-years; their 95% intervals, corrected for the overlap between the resampled training sets (Nadeau & Bengio, 2003), are narrow (AUC widths below 0.03), so the expectation models are estimated precisely.

Third, feature selection. Features are not hand-curated. A pre-specified core of fine-grained game-state fields (down, distance, field position, score margin, clock, and timeouts) is always retained; the remaining context variables (Table 4, Supplementary material) are chosen by stability selection (Meinshausen & Bühlmann, 2010). Over B subsamples of team-year groups, letting $\hat{S}_K^{(b)}$ be the K highest-importance features in subsample b, the selection frequency of feature k is

$$\hat{\Pi}_k^{(K)} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left\{ k \in \hat{S}_K^{(b)} \right\}, \tag{2}$$

and we retain features with $\hat{\Pi}_k^{(K)} \geq \pi$ at threshold $\pi = 0.6$ (taking the union over shortlist lengths K), which prunes collinear inputs without materially changing accuracy.

2.3 Contemporary-group (era) adjustment

Before the transmission (C1), repeatability (C2), and selection (S) analyses (not the gene-fitting step above), each standardized phenotype is re-expressed as a deviation from its own season's league mean (within-season demeaning),

$$\tilde{g}_c^{\,\mathrm{adj}} = \tilde{g}_c - \frac{1}{|C_t|}\sum_{c' \in C_t} \tilde{g}_{c'}, \qquad t = \mathrm{season}(c), \tag{3}$$

where $\tilde{g}_c^{\,\mathrm{adj}}$ is the era-adjusted phenotype, $\tilde{g}_c$ the standardized phenotype from above, C_t the set of all coach-seasons in season t, $|C_t|$ its size, and c' the index running over those coach-seasons. Because both a mentor's and a protege's values are re-centered on their own seasons, league-wide drift is removed exactly once and is not counted as transmission or selection. This is the single era control, the cultural analog of the contemporary-group (herd-year-season) adjustment used to estimate breeding values in the animal model; no further season or era term enters the models. Coach WAR is already a within-season, replacement-relative quantity (about 3% between-season variance) and receives no further adjustment.

2.4 Reliability and measurement error

Each coach-season phenotype is an estimate with a known sampling variance. For a probability sub-trait, a coach-season residual has sampling variance $v_i = \sum_j \hat{p}_j(1-\hat{p}_j)/n_i^2$ over its n_i plays, where $\hat{p}_j$ is the model's predicted probability on play j; for a regression sub-trait the analogous quantity is $v_i = \hat{\sigma}^2_{r,i}/n_i$, where $\hat{\sigma}^2_{r,i}$ is the variance of the play-level residuals $a_{cj} - \hat{a}_{cj}$ for that coach-season. We estimate the between-coach-season variance $\hat{\tau}^2$ with the DerSimonian–Laird random-effects estimator (DerSimonian & Laird, 1986). Each coach-season enters its composite with weight equal to its reliability,

$$\mathrm{rel}_i = \frac{\hat{\tau}^2}{\hat{\tau}^2 + v_i}, \tag{4}$$

so a component measured from few plays contributes less. The single-component shotgun gene has no siblings to weight against, so the same reliability is used to shrink each coach-season toward the population mean,

$$g_i^{\mathrm{shrunk}} = \bar{g} + \mathrm{rel}_i\,(g_i - \bar{g}), \tag{5}$$

before standardization, where $\bar{g}$ is the mean gene across coach-seasons. The same per-observation sampling variance, rescaled to the standardized scale, is carried explicitly as a measurement-error level in the heritability (C1) and repeatability (C2) models, so both are estimated for the latent trait rather than its noisy observation, and composites propagate their components' sampling variance. Reliabilities vary across sub-traits (for example the offensive-aggression components run from about 0.64 for fourth-down to 0.91 for pass-heavy), which is why carrying measurement error, rather than treating every phenotype as equally precise, matters for the cross-trait comparison in C3.

2.5 C1: transmissibility (h^2)

A coordinator-to-head-coach transition pairs two different coaches. The outcome is the protege's own phenotype once they became a head coach; the predictor is the phenotype of the head coach they apprenticed under, measured on that mentor's team over the seasons the protege served there as a coordinator. Each coach contributes at most one transition per coordinator role (the most recent coordinator stint preceding a head-coaching stint), which avoids pseudo-replication. The comparison is therefore between two different people, the mentor and the protege.

We estimate h^2 with a Bayesian multilevel parent-offspring model fit at the coach-season grain (Nakagawa & Schielzeth, 2010; Wilson et al., 2010). For protege i, a latent mentor level m_i and a latent own level a_i are linked by a regression with intercept α, transmission slope β, and residual standard deviation σ_a,

$$a_i = \alpha + \beta\, m_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma_a), \tag{6}$$

and each individual mentor-overlap season x_{ij} and protege head-coaching season y_{ik} loads on the corresponding latent level with a variance that combines a shared true within-coach season-to-season component τ^2 and the known measurement variance of that observation:

$$x_{ij} \sim \text{Normal}\big(m_i,\ \sqrt{\tau^2 + \text{se}_{ij}^2}\big), \qquad y_{ik} \sim \text{Normal}\big(a_i,\ \sqrt{\tau^2 + \text{se}_{ik}^2}\big). \tag{7}$$

Modeling the individual seasons, rather than collapsing each coach to a single average first, is what makes this an errors-in-variables estimator. All of a mentor's overlap seasons $\{x_{ij}\}$ load on the one latent mentor level m_i, and all of a protege's head-coaching seasons $\{y_{ik}\}$ load on the one latent own level a_i; the slope β then relates these two latent levels, not the raw season averages. A multi-season overlap and a multi-season head-coaching stint are connected through this shared latent level, one per coach, rather than by matching seasons to seasons. The difference from pre-averaging is that m_i and a_i are inferred with their own uncertainty: a season measured from few plays has a large se and moves its latent level less, so noisy seasons are down-weighted rather than counted equally, and the mentor predictor enters carrying its measurement variance. Plugging in a plain season average would instead discard that precision and, because the predictor is measured with error, attenuate the slope toward zero. The transmissibility is the parent-offspring regression slope at the coach-season grain,

$$h^2 \;=\; \frac{\beta\, \sigma_m^2}{\sigma_m^2 + \tau^2} \;=\; \beta \times \text{repeatability}, \tag{8}$$

where σ_m^2 is the variance of the latent mentor levels across mentors, so h^2 is bounded above by the mentor repeatability $\sigma_m^2/(\sigma_m^2 + \tau^2)$: a trait cannot transmit more faithfully than it is a stable trait to begin with. Because a mentor transmits a whole style rather than half a genome, the slope itself is the transmissibility, with no Mendelian doubling. Priors are Normal(0, 1) on the intercept and slope (phenotypes are z-scored) and half-Normal(0, 1) on the variance components; sampling (in PyMC) uses four chains with 2000 post-warmup draws each, a non-centered parameterization, and target diagnostics $\hat{R} \leq 1.01$ and bulk effective sample size > 400 (met for every fit). Two robustness refits are reported alongside the primary slope in the Results: a Bayesian variant that adds a mentor-level

varying intercept (for mentors with several proteges) and a frequentist parent-offspring slope with mentor-clustered standard errors.

2.6 C2: repeatability

Per-trait repeatability is a Bayesian variance-components intraclass correlation on the full head-coach coach-season panel, with the per-season measurement variance carried as an explicit level so the ratio is the latent-trait repeatability, $\sigma^2_{\text{between}}/(\sigma^2_{\text{between}} + \sigma^2_{\text{within}})$. It is the ceiling for h^2. This ceiling is not imposed on the sampler: the slope β in Equation 8 carries a Normal$(0, 1)$ prior and is free to exceed one, so h^2 is free to exceed the repeatability, and the bound is an empirical property of the estimates rather than a constraint. Because this repeatability is fit by a separate model on the full head-coach panel, its posterior and the h^2 posterior are independent and can cross (as they do, at the median, for the traits that transmit near their ceiling); we therefore report $P(h^2 >$ repeatability) per trait, where a value near one would flag a genuine specification problem rather than an artifact of shared parameters. The repeatable-but-not-heritable finding for offensive aggression rests on β (and hence h^2) being indistinguishable from zero, not on the ceiling.

2.7 Selection (S)

Coach WAR is our fitness proxy, and it is chosen deliberately. It estimates the wins a head coach adds relative to a replacement-level alternative, adjusting for roster talent (including injuries) and strength of schedule, so it credits the coaching contribution rather than a strong roster; it is validated out of sample on held-out coaches in companion work (Anonymous, 2026). Raw win rate, the payoff signal used to gauge success in the predecessor study (Mesoudi, 2020), confounds a coach's contribution with roster strength and schedule; WAR adjusts for those, addressing a limitation that study noted in win rate. It cannot remove single-season luck, which we handle separately through inverse-variance weighting (below). Being replacement-relative and computed within each season, it is a relative-fitness analog centered near zero (median approximately zero) and nearly era-flat (about 3% between-season variance), so, unlike the genes, it needs no separate era adjustment. WAR is taken from the companion estimator at a fixed pinned commit (`78267b5`). The preregistration pinned commit `dffe2f1`, whose trajectory file still carried WAR estimated before the companion project corrected an imputation leak; as the preregistration specifies for a revised WAR, we report S on the corrected estimates. On the preregistered file every S is within 0.03 of the values reported here and no hypothesis verdict changes; the one interpretive difference is shotgun, whose S there is 0.08 ($[-0.03, 0.17]$).

Selection S is then the era-adjusted, inverse-variance-weighted correlation of a trait with Coach WAR: within-season demean the standardized gene and WAR, weight each coach-season by its games-based WAR precision, and take the coach-clustered weighted correlation. Single-season WAR is noisy, and because mean-zero noise in the outcome attenuates a correlation rather than inflating it, this estimate is a conservative floor.

2.8 C3 and decision rules

For the ten sub-traits we assemble the pair (h^2, S) and compute their Spearman rank correlation with a 95% bootstrap interval that resamples the traits and draws h^2 from its posterior and S from its bootstrap. The rank form is the established test of the heritability-fitness relationship in quantitative genetics (Kruuk et al., 2000; Mousseau & Roff, 1987). Each hypothesis is judged by one pre-specified interval rule. H1 is supported if the approach family's mean S exceeds the identity family's and the identity family's mean h^2 exceeds the approach family's, each difference with a 95% interval excluding zero. H2 is supported if (aggression repeatability minus aggression h^2) has a 95% interval excluding zero. H3 is supported if the Spearman correlation is negative with a 95% bootstrap interval excluding zero.

2.9 Exploratory analyses

We report three analyses outside the confirmatory protocol. The first two, a within-era Moran's I on the mentor network and a reproductive-fitness rate, are corrected among themselves under a Benjamini–Hochberg false-discovery rate (Benjamini & Hochberg, 1995), separately from the confirmatory tests; the third, horizontal transmission, is a Bayesian model reported by posterior intervals and is described last. The Moran's I is computed on the mentor network (an undirected, row-normalized adjacency of mentor–protege ties, with a permutation null that shuffles trait labels only within career-midpoint era blocks) (Moran, 1950). Reproductive fitness is an exposure-normalized rate: head-coaching offspring (proteges who themselves became head coaches) per head-coaching season, with a minimum-exposure floor, together with the test of whether coach quality predicts that rate net of exposure. The corrected family is exactly these two analyses, five tests in all: one Moran's I per gene (four) and the reproductive-rate quality test. The remaining reproductive-fitness quantities (the distribution and the raw correlations of offspring count with career WAR and with tenure) are reported as descriptive effect sizes, not as corrected hypothesis tests.

The third exploratory analysis measures horizontal (peer-to-peer) transmission, the complement of the vertical mentor–protege channel. Because horizontal diffusion lives in the season-to-season drift of the league, here we use the absolute standardized phenotype (z-scored over the trait window but not contemporary-group demeaned), so the drift the confirmatory analyses remove as a confounder is retained as the signal. For each trait we ask whether a head coach's next-season phenotype is drawn toward the current league level, beyond his own previous value and his own baseline, through a Bayesian multilevel autoregression $z_{i,t+1} = a + u_i + \phi\, z_{i,t} + \kappa\, \bar{z}_{-i,t} + \varepsilon$, where $\bar{z}_{-i,t}$ is the leave-one-out league mean in season t and u_i a coach varying intercept carrying the coach's own baseline. A convergence coefficient $\kappa > 0$ is the signature of frequency-biased (conformist) social learning, copying what is common, and it uses no WAR. The ten sub-traits are fit jointly with κ partially pooled across traits ($\kappa_p \sim \text{Normal}(\mu_\kappa, \sigma_\kappa)$), so the population estimate μ_κ and the cross-trait shrinkage regularize the per-trait estimates in place of a Benjamini–Hochberg step; the four gene-level composites are fit singly. This is a separate family from the network and reproductive tests, reported by posterior HDI and $P(\kappa > 0)$ rather than corrected p-values. As a robustness check we add a per-trait linear season trend; because conformity and the linear drift it produces are nearly collinear, that fit is a conservative floor rather than a clean separation.

3. Results

3.1 The phenotypes are repeatable coach traits

Before asking how a trait transmits or whether it wins, we confirm it is a stable individual characteristic rather than a team artifact. Every trait is repeatable: the latent-trait repeatabilities (Table 3) range from 0.34 (pass-heavy) to 0.58 (fourth-down), all with 95% intervals excluding zero, so each phenotype carries substantial stable between-coach signal net of measurement noise. This is the quantitative-genetics prerequisite for a meaningful heritability: a trait must be repeatable before it can be transmissible.

Three companion analyses corroborate that the genes belong to the coach rather than the team or its roster (Table 2). First, a one-way variance decomposition attributes far more of each gene's variance to coach identity than to team identity (for composite offensive aggression, 0.50 versus 0.14). Because η^2 inflates mechanically with the number of groups, and a coach maps to far more levels than a team, we benchmark each against a cardinality-matched label-permutation null: the coach excess over its null is about 0.29 against 0.08 for team, so the coach signal is roughly three times larger, and the same ordering holds for every gene. Second, when coaches change teams their tendencies travel with them: across the 32 coaches who moved, composite offensive aggression at the new team tracks the gene they showed at the old one ($r = 0.55$, 95% CI [0.29, 0.73], on the era-adjusted gene), and in a mover regression the coach's prior-team gene carries $\beta = 0.54$ ([0.26, 0.88]) net of the destination's

Table 2. The offensive genes are coach traits. One-way variance shares (coach, team, season; marginal η^2, which need not sum to one because coach and team are confounded), cross-team travel (era-adjusted across-team correlation among the 32 movers, 95% CI), and year-to-year persistence (lag-1 correlation between consecutive coach-seasons; computed for the aggression genes).

Gene	coach η^2	team η^2	season η^2	travel r [95% CI]	persist. r
Composite offensive aggression	0.50	0.14	0.18	0.55 $[0.29, 0.73]$	0.50
fourth-down	0.51	0.07	0.26	0.09 $[-0.24, 0.42]$	0.53
pass-heavy	0.49	0.18	0.08	0.32 $[-0.03, 0.60]$	0.55
deep-pass	0.44	0.12	0.05	0.17 $[-0.20, 0.51]$	0.46
two-point	0.44	0.09	0.17	0.30 $[0.04, 0.60]$	0.43
Shotgun	0.57	0.12	0.48	0.28 $[-0.18, 0.68]$	–
Tempo	0.52	0.11	0.07	0.48 $[-0.11, 0.82]$	–

pre-arrival identity. Third, the genes persist from season to season (composite offensive aggression lag-1 $r = 0.50$, with its four components between 0.43 and 0.55). At only 32 movers the travel test is imprecise for any single gene, which is why the composite, which averages out component sampling noise, travels most cleanly.

Together with the repeatabilities, these lines establish that the phenotypes are stable properties of the coach rather than artifacts of a roster, the prerequisite for treating them as heritable-eligible traits.

3.2 Transmissibility and selection come apart

Table 3 reports, for each trait, the preregistered transmissibility (h^2, C1), repeatability (C2), and selection (S). Two facts stand out. First, transmissibility varies sharply across traits that are all similarly repeatable: the schematic-identity genes transmit strongly (shotgun $h^2 = 0.45$, 95% HDI $[0.27, 0.65]$; no-huddle 0.30, $[0.09, 0.51]$) as does defensive aggression (0.49, $[0.13, 0.85]$), while offensive aggression does not transmit at all (-0.15, $[-0.46, 0.12]$). Second, selection follows a nearly opposite ordering: the approach components predict WAR (offensive aggression $S = 0.18$, $[0.08, 0.27]$; defensive aggression 0.17, $[0.04, 0.29]$; pass-heavy 0.17; pass-rush 0.19) while the identity genes predict it weakly or not at all (shotgun 0.10, $[0.00, 0.20]$; tempo 0.02). The frequentist mentor-clustered slopes and the mentor varying-intercept refits agree with the Bayesian h^2 (for example shotgun clustered slope 0.47, $p < 0.001$; defensive 0.61, $p = 0.03$; offensive aggression -0.03, $p = 0.84$). Figure 1 plots each trait's h^2 against its repeatability ceiling and makes the pattern legible at a glance.

3.3 The heritability by selection map and the preregistered verdicts

Placing the four composites in $h^2 \times S$ space (Figure 2) makes the structure visible. Offensive aggression sits in the adaptive-but-personal corner (selected, not heritable); tempo sits in the heritable-but-neutral corner (transmitted, not selected), with shotgun beside it, strongly transmitted but at most weakly selected; defensive aggression sits clearly in the adaptive-and-heritable corner. The two axes are close to orthogonal.

The three preregistered decision rules resolve as follows.

H2 is supported. Offensive aggression is repeatable (0.40) but not heritable ($h^2 = -0.15$); the preregistered contrast, repeatability minus h^2, is 0.55 with a 95% HDI of $[0.24, 0.87]$, excluding zero, and $P(h^2 >$ repeatability$) = 0.00$. Offensive aggression is a real, stable, personal trait that does not pass down the tree. The contrast behaves as the prereg intended elsewhere too: for the traits that transmit near their ceiling (shotgun, defensive aggression, box-stacking, pass-rush) h^2 and repeatability are statistically indistinguishable ($P(h^2 >$ repeatability$)$ between 0.5 and 0.75, each h^2

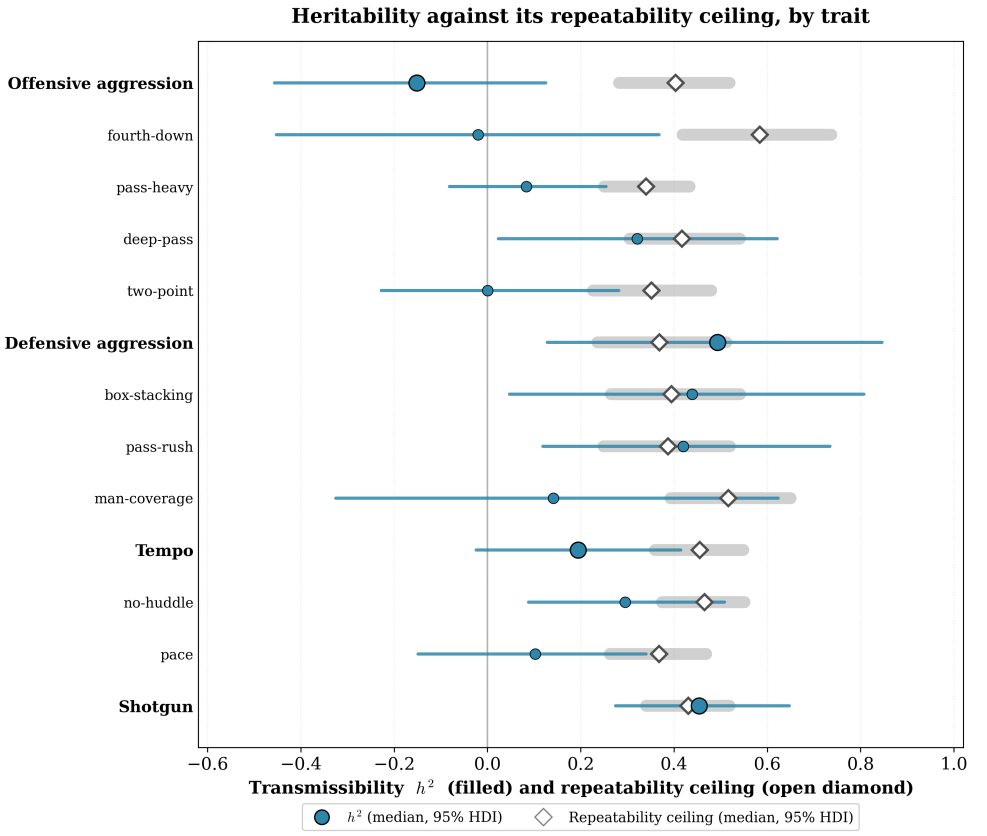


Figure 1. Transmissibility (h^2, filled points with 95% HDIs) against its repeatability ceiling (open diamond, median; gray band, repeatability 95% HDI) for every trait, grouped by gene and colored by family. A heritability cannot exceed its repeatability in the population; because h^2 and repeatability are estimated from separate models with wide posteriors, their medians can cross for traits that transmit near the ceiling, but every h^2 point falls within the repeatability's credible interval (the posterior contrast $P(h^2 >$ repeatability) is 0.5 to 0.75 for these, that is $h^2 \approx$ repeatability). Offensive aggression and its components are repeatable but have h^2 indistinguishable from zero, a clean gap ($P = 0.00$), the preregistered repeatable-but-not-heritable pattern (H2); schematic identity and the defensive front transmit near their ceiling.

inside the repeatability's credible interval), so the apparent crossings in Figure 1 are estimation noise near the ceiling, not violations of the bound.

H1 is not supported as a conjunction, though its selection half holds. The approach family is more selected than the identity family (mean-S difference +0.11, 95% interval $[0.004, 0.22]$, excluding zero), but the identity family is not reliably more heritable than the approach family (mean-h^2 difference +0.15, $[-0.12, 0.43]$, including zero). The heritability half fails for a substantive reason: defensive aggression is an approach trait that is nonetheless highly heritable ($h^2 = 0.49$), so the approach family does not have uniformly low transmissibility. Whether defensive aggression is really a pure approach trait is itself open to question: on defense the aggression measure blends a risk posture (loading the box, sending rushers) with a scheme choice (man versus zone coverage), so its high transmissibility may be the schematic half showing through rather than a counterexample to the taxonomy. We take up this classification directly in Section 4.3.

H3 is inconclusive. Across the ten sub-traits the Spearman correlation between h^2 and S is +0.31, but with a 95% bootstrap interval of $[-0.52, 0.72]$ (Figure 3) that runs from a strong negative to a strong positive slope. The ten trait points form a broad cloud rather than a line, so many slopes

Table 3. Transmissibility (h^2, C1), repeatability (C2), and selection (S) for the four composites and the ten sub-traits. h^2 and repeatability are posterior medians with 95% HDIs; S is the era-adjusted inverse-variance-weighted correlation with WAR with a 95% coach bootstrap CI. All C1/C2 fits met $\hat{R} \leq 1.01$ and bulk ESS > 400.

Trait	h^2	repeatability	S
Offensive aggression	−0.15 [−0.46, 0.12]	0.40 [0.28, 0.52]	0.18 [0.08, 0.27]
fourth-down	−0.02 [−0.45, 0.37]	0.58 [0.42, 0.74]	0.05 [−0.05, 0.13]
pass-heavy	0.08 [−0.08, 0.25]	0.34 [0.25, 0.43]	0.17 [0.05, 0.28]
deep-pass	0.32 [0.02, 0.62]	0.42 [0.30, 0.54]	0.08 [−0.01, 0.17]
two-point	0.00 [−0.23, 0.28]	0.35 [0.23, 0.48]	−0.02 [−0.12, 0.07]
Tempo	0.19 [−0.02, 0.41]	0.45 [0.36, 0.55]	0.02 [−0.07, 0.13]
no-huddle	0.30 [0.09, 0.51]	0.47 [0.37, 0.55]	−0.03 [−0.11, 0.07]
pace	0.10 [−0.15, 0.34]	0.37 [0.26, 0.47]	0.06 [−0.05, 0.18]
Defensive aggression	0.49 [0.13, 0.85]	0.37 [0.24, 0.51]	0.17 [0.04, 0.29]
box-stacking	0.44 [0.05, 0.81]	0.39 [0.26, 0.54]	0.06 [−0.05, 0.17]
pass-rush	0.42 [0.12, 0.73]	0.39 [0.25, 0.52]	0.19 [0.06, 0.31]
man-coverage	0.14 [−0.33, 0.62]	0.52 [0.39, 0.65]	0.14 [−0.01, 0.29]
Shotgun	0.45 [0.27, 0.65]	0.43 [0.34, 0.52]	0.10 [0.00, 0.20]

fit about equally well and the sign is not identified: a small change in the leverage of any one trait would flip it, from the negative trade-off implied if offensive aggression dominates to the positive relationship implied by the other three genes. With so few traits the cross-trait test is underpowered, so we read the result as uninformative about the sign rather than as evidence for or against a trade-off.

Taken together, the verdicts describe a *decoupling* rather than a trade-off. Transmission and selection are close to independent across coaching traits, with two archetypes, inherited identity that is at most weakly selected (tempo, and shotgun at the margin) and selected-but-personal offensive aggression, and one trait, defensive aggression, that clearly occupies the both corner and is chiefly responsible for the absence of a clean negative relationship.

3.4 Exploratory: network signal, horizontal transmission, and reproductive success

Three exploratory analyses round out the picture; none is confirmatory. The first two (network signal, reproductive rate) are corrected among themselves under a Benjamini–Hochberg false-discovery rate; the third (horizontal transmission) is a Bayesian model reported by posterior intervals, a separate family.

First, phylogenetic signal on the mentor network (Table 5, Supplementary material). Moran's I, the network analog of the tendency of related individuals to resemble one another, is positive for the schematic and defensive genes and essentially zero for offensive aggression (defensive aggression $I = 0.20$, within-era permutation $p = 0.002$; tempo 0.12, $p = 0.008$; shotgun 0.08, $p = 0.04$; offensive aggression -0.01, $p = 0.52$). Under the exploratory correction the defensive and tempo signals survive. The pattern mirrors the transmission results exactly: network-adjacent coaches resemble one another in the traits that transmit and not in the trait that does not, a second and independent line of evidence for the same decoupling.

Second, horizontal transmission, measured rather than inferred (Figure 4). Holding a coach's own previous value and baseline fixed, head coaches converge each season on the prevailing league level: pooled across the ten sub-traits the convergence coefficient is $\mu_\kappa = 0.22$ (95% HDI [0.02, 0.40], $P(\kappa > 0) = 0.98$). The pull is sharply selective and falls on exactly the behaviors that swept the league: fourth-down aggression ($\kappa = 0.57$, HDI [0.40, 0.75]), two-point attempts (0.44, [0.24, 0.64]), deep passing (0.31, [0.04, 0.60]), the offensive-aggression composite (0.37, [0.16, 0.57]), and shotgun (0.20,

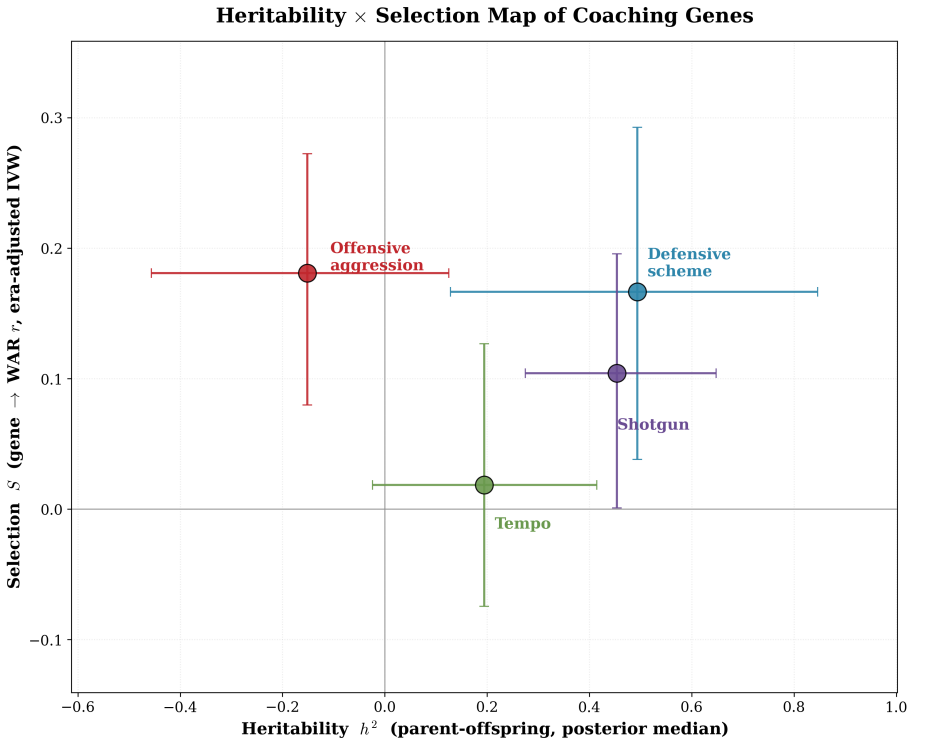


Figure 2. Heritability (h^2, C1 posterior median) versus selection (S, era-adjusted inverse-variance-weighted gene-WAR correlation) for the four gene-level composites, with 95% intervals on both axes. Offensive aggression is adaptive but personal; tempo is heritable but neutral, and shotgun heritable and at most weakly selected; defensive aggression is both. The axes are nearly independent.

[0.09, 0.30]) all show credible convergence, while pace, no-huddle, pass-heavy, and the defensive traits do not. This is the horizontal channel made visible: fourth-down aggression, whose vertical h^2 is indistinguishable from zero, is nonetheless pulled hard toward the league norm, the clearest case of a trait that spreads across contemporaries rather than down lineages. Because conformity and the drift it produces are nearly collinear, a model that also absorbs a per-trait linear trend shrinks the pooled estimate to 0.05 ([−0.10, 0.21]); we therefore read κ as evidence that coaches converge on common norms, consistent with frequency-biased social learning, while noting that convergence by peer imitation and by a shared response to a changing environment are not fully separable here.

Third, reproductive success, the other face of fitness (Figure 5). Counting a mentor's distinct proteges who themselves became head coaches, the distribution is heavy-tailed: across 265 modern-era head coaches (first head-coaching season in 1970 or later, matching the post-merger era) the Gini coefficient is 0.56 and the most prolific decile produced 35% of all head-coaching descendants, with Marty Schottenheimer and Bill Parcells (13 each), Bill Belichick (12), and Andy Reid (11) at the top. Reproductive success tracks career WAR (r = 0.35), but the link runs through longevity rather than per-season quality: offspring count correlates 0.74 with seasons coached, and net of tenure the exposure-normalized rate (offspring per head-coaching season) is essentially unrelated to quality (partial r = 0.07). A coach's ecological fitness (winning) and reproductive fitness (descendants) are thus only weakly linked, and through survival: coaches shape the league less by winning in any given season than by lasting long enough to send many proteges into head-coaching chairs, carrying their schematic identity, though not their competitive edge, with them.

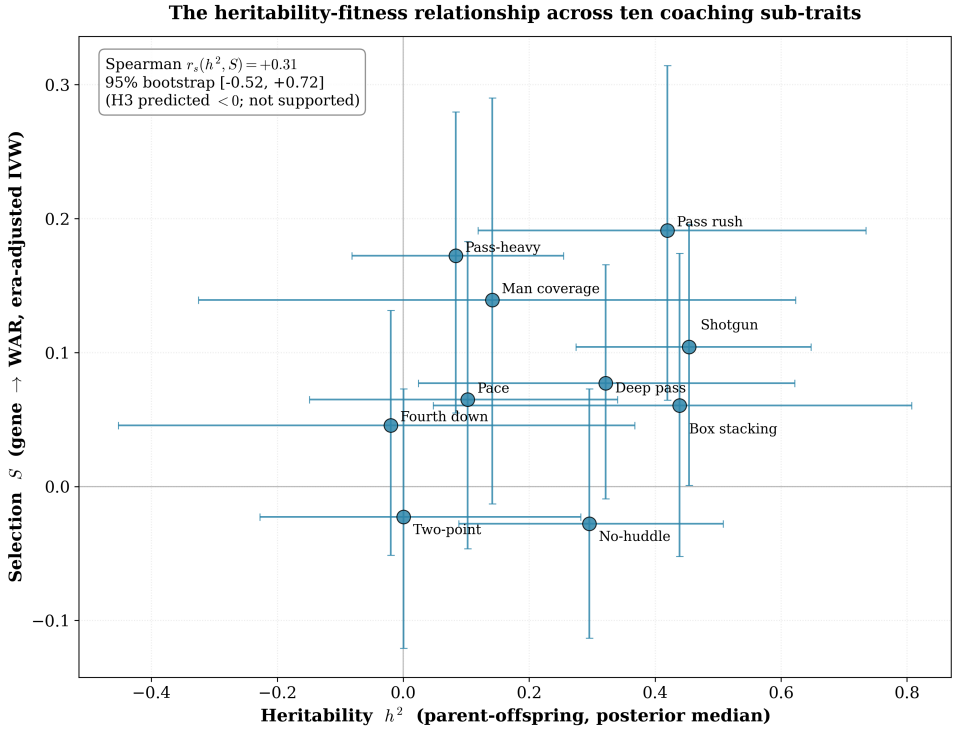


Figure 3. The cross-trait heritability-fitness relationship, the cultural analog of the wild-population test of Mousseau and Roff (1987) and Kruuk et al. (2000): each of the ten sub-traits placed by transmissibility h^2 and selection S, with 95% intervals (a single colour throughout; the approach-versus-identity contrast is drawn in the text, not by segmenting the figure). The Spearman rank correlation (H3) is +0.31 ($[-0.52, 0.72]$); the interval is too wide to fix the sign, so the cross-trait trade-off test is inconclusive. The decoupling we describe rests on where the traits sit in $h^2 \times S$ space (Figure 2), not on this slope.

4. Discussion

4.1 Decoupling, not a trade-off

We set out to test whether the traits that transmit down a coaching lineage are the traits that win, expecting the negative heritability–fitness relationship familiar from biological populations. That specific trade-off did not appear, but neither did any clear relationship in the other direction: with so few traits the h^2-by-S association is indeterminate (cross-trait Spearman +0.31 across the ten sub-traits, interval spanning zero). The traits scatter in a loose square rather than along a line, and the estimate is held back from being negative chiefly by defensive aggression, the trait that sits most clearly in the both-corner. We therefore do not read the cross-trait slope as a structure in its own right. What is interpretable is where individual traits fall in the h^2-by-S plane, and there two archetypes stand out. Schematic-identity traits, shotgun and tempo, are transmitted faithfully (h^2 up to 0.45) yet carry little fitness consequence (tempo S = 0.02; shotgun S = 0.10, an interval that only just excludes zero): in evolutionary terms they behave like heritable, nearly neutral markers (Kimura, 1968), tags of lineage that ride along with apprenticeship rather than with winning. Offensive aggression is the mirror image, a repeatable and selected but non-transmitted trait (h^2 indistinguishable from zero, S = 0.18): among the offensive traits we measured, the one that predicts success, in-game aggression, is a personal characteristic that does not pass down. This last pattern is the preregistered H2, and it is the sharpest single result: a trait can be a stable individual property (repeatability 0.40) and predict success and still not be inherited.

We preregistered a negative relationship and did not find one. That null invites a sharper reading

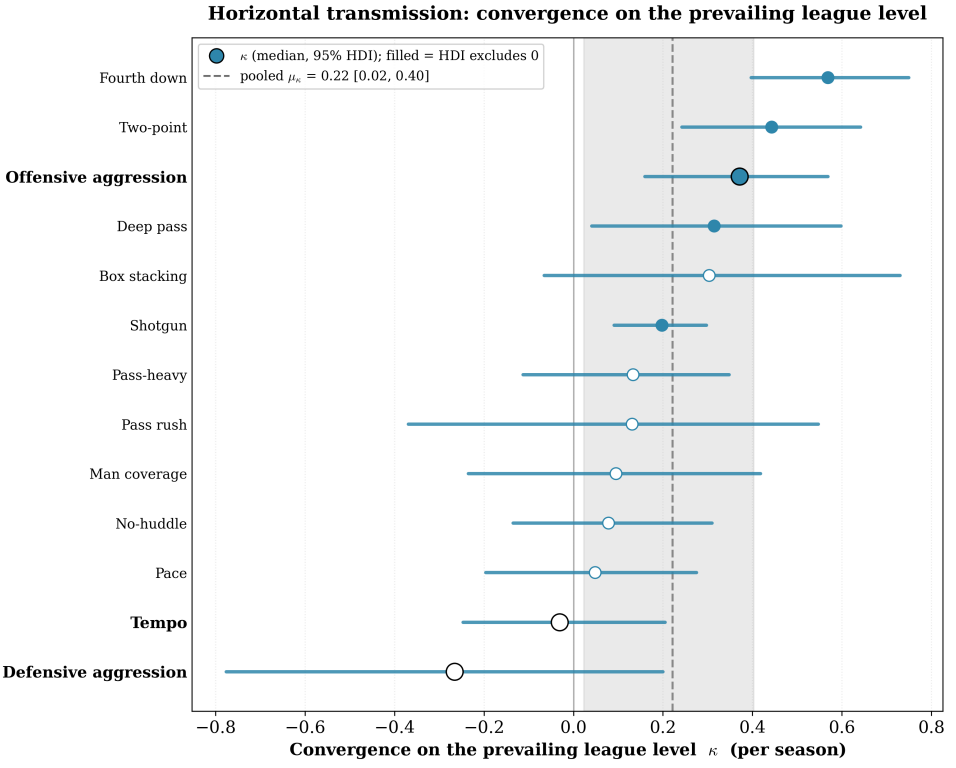


Figure 4. Horizontal transmission: per-trait convergence on the prevailing league level (κ, posterior median and 95% HDI) against the pooled estimate μ_κ (dashed line and band). Filled markers have an HDI excluding zero. Convergence is credible and largest for the behaviors that diffused across the league (fourth-down, two-point, deep passing, the offensive-aggression composite, shotgun) and absent for the traits that did not. The absolute (non-era-adjusted) phenotype is used here so the league-wide drift is retained as the horizontal signal.

through the breeder's equation, $R = h^2S$, which gives a trait's per-generation response to selection along a lineage. For offensive aggression the lineage prediction is essentially nil ($R \approx h^2S \approx 0 \times 0.18 = 0$), because the trait is not transmitted, and yet its league-wide mean rose steadily over our window. A quantity that the breeder's equation says should barely change down the tree changed markedly across the population, the empirical signature that a second, non-vertical channel is at work.

4.2 Two channels: vertical down the tree, horizontal across the league

The classical trade-off assumes a single inheritance channel along which selection erodes heritable variance. Coaching has two such channels (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981). The schematic-identity traits spread *vertically*, down the offensive-coordinator apprenticeship. Shotgun shows this most clearly, with a parent-offspring h^2 of 0.45 whose interval excludes zero; tempo points the same way, with a positive h^2 (0.19) and a within-era network signal that survives correction (Table 5). Offensive aggression, lacking a vertical channel ($h^2 \approx 0$, no network autocorrelation), spread *horizontally* instead. Its league-wide mean rose monotonically across our window (linear trend $r = 0.92$ with season, $p < 0.001$; a large early-to-late shift, Cohen's $d = 6.6$), crossing from below the time-invariant, season-agnostic baseline in the late 2000s to clearly above it by the early 2020s: the classic profile of a payoff-driven innovation copied among contemporaries. Because the expectation model excludes season, that drift is genuine change in behavior rather than a modeling artifact; with no vertical pathway to carry it, that drift is horizontal spread among

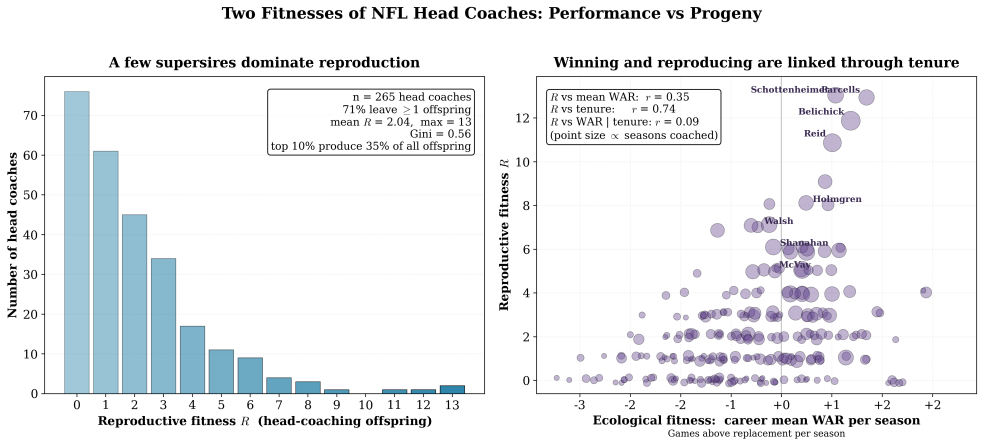


Figure 5. Two fitnesses of NFL head coaches. Left: the distribution of reproductive fitness (number of head-coaching proteges) is heavy-tailed (Gini 0.56; top decile produces 35% of all descendants). Right: ecological fitness (career mean WAR per season) versus reproductive fitness, point size proportional to seasons coached; the positive raw association ($r = 0.35$) is carried by tenure (offspring versus seasons coached $r = 0.74$), and net of tenure quality barely predicts offspring (partial $r = 0.09$).

contemporaries, which we now measure rather than only infer. Holding a coach's own history fixed, head coaches converge each season on the prevailing league level, credibly so for fourth-down aggression and the other behaviors that swept the league (Figure 4); fourth-down aggression, with h^2 at zero, is pulled hardest of all. That convergence is the footprint of frequency-biased copying, though imitation and a shared response to a changing environment cannot be fully separated in it.

This is where the study answers the questions its predecessor left open. Mesoudi (2020) showed that football managers choose formations by strategic social learning, weighting their own and the wider population's recent use of a tactic, so that a successful formation diffuses horizontally as managers copy it from their peers; and he named two channels his data could not reach, transmission down mentor–protege lineages and the flow of influence along the network ties between managers, illustrated by the lineage of managers who carry Bielsa's approach. Our design adds the channel his could not reach. The mentor–protege h^2 slopes, and the Moran's I on the same apprenticeship network (Table 5), put numbers on the lineage transmission he pointed to but could not measure; both, however, describe the vertical channel, not the peer ties he named. The horizontal channel is more his than ours: he modeled peer copying directly, from formation-adoption dynamics, whereas we measure only its aggregate footprint, the season-by-season convergence of coaches on the prevailing norm (Figure 4), not the dyadic copying itself. With that asymmetry stated, the two studies interlock. The trait Mesoudi's horizontal lens tracks, the fitness-linked one, is exactly the trait our vertical lens finds absent from the coaching tree; the trait our vertical lens finds transmitting is not the one that sweeps the league on a fitness gradient. Vertical transmission carries system identity; horizontal transmission carries the winning tactic. A study looking only down the tree would conclude aggression is absent from coaching culture; a study looking only at league-wide diffusion would miss that scheme is inherited. Read together, the two channels partition the strategic repertoire, and the era structure we remove as a statistical confounder is, substantively, the horizontal channel itself.

4.3 Defensive aggression: attitude and scheme entangled

Defensive aggression is the composite that most clearly both transmits ($h^2 = 0.49$) and predicts winning ($S = 0.17$), and it is chiefly why the cross-trait relationship is not cleanly negative. It is best

read not as a counterexample but as a reminder that our two-family taxonomy, approach versus identity, is cleaner on offense than on defense. "Aggression" on defense bundles dimensions the offensive split holds apart, and they differ in how schematic they are. The number of pass rushers, for one, is as much a schematic property as an attitudinal one: whether a front rushes four or five is largely a matter of running a 4–3 or a 3–4 base, not a play-by-play risk decision, and box-stacking is similarly tied to front and personnel. Man versus zone is the least schematic of the three, closer to a situational, matchup-driven call. Transmissibility follows that ordering: box-stacking and pass-rush transmit strongly (h^2 = 0.44 and 0.42), like schematic identity, while man coverage transmits weakly (h^2 = 0.14). A composite that folds these together inherits transmissibility from its schematic components and a fitness link from its aggressive posture, which is precisely the both-corner position it occupies. The organizing distinction is attitude versus scheme rather than offense versus defense, and on defense the two are entangled within a single composite; a cleaner future decomposition would separate a defense's situational risk posture from its base front and coverage system.

4.4 Coaching lineages transmit systems, not success

A common rationale for hiring from established coaching trees is that success flows from mentor to protege. That claim finds little support here: working under a successful mentor does not predict a protege's own success (mentor-protege WAR resemblance r = 0.11, 95% CI [−0.04, 0.25], mentor-clustered p = 0.14; n = 193 pairs across 88 mentors, not distinguishable from zero), and the in-game risk tendency that does predict success is not passed down. What coaching trees reliably transmit is schematic system, formation and tempo identity, and specifically through the coordinator who would install it. There is a labor-market reading of this asymmetry: an organization can observe and verify the system a coordinator runs far more readily than whether he will win, so hiring on visible schematic pedigree is hiring on the trait that transmits, not the trait that drives value. What is legible (system) and what is decisive (in-game approach) come apart, and only the former leaves a lineage to point to. We offer this as interpretation rather than a tested claim; the confirmatory result is the decoupling itself.

4.5 Limitations

The transmission samples are small (40 offensive and 17 defensive coordinator-to-head-coach transitions, and only 12 for man coverage), so the h^2 posteriors are wide and the cross-trait test in particular has low power; H3's wide interval should be read as inconclusive about a weak relationship, not as evidence for a positive one. Coaching staffs are groups, but our transmission model reduces each to the single head-coach-to-coordinator (OC or DC) link we can measure; position coaches, multiple simultaneous coordinators, and overlapping staffs, the real structure of a coaching tree, are simplified away. The genes attribute team play-calling to the head coach and do not isolate a coordinator's individual contribution; the transmission tests therefore ask whether the head-coaching style a coach apprenticed under predicts his own later style, a mentor-protege resemblance that is consistent with social learning but cannot, on its own, be separated from assortative hiring, in which head coaches recruit coordinators who already share their approach. Distinguishing the two would take a phenotype measured before the apprenticeship, or staffs assigned rather than chosen; neither is generally available, since a coach's play-calling style is not defined until he runs a unit. We therefore read h^2 as vertical transmissibility in the broad sense. The defensive-aggression measures begin in 2016 and man coverage in 2018, and the defensive-aggression to WAR link may partly reflect reverse causation (better defensive talent enables man coverage and extra rushers). Finally, single-season WAR is noisy; we address this with inverse-variance weighting and treat the season-level estimates as conservative floors. One asymmetry in evidential status follows from the design: our mentor network encodes documented apprenticeship ties rather than the full web of peer influence among contemporaries, so the vertical channel is estimated directly, from both measured endpoints, while

the horizontal channel is measured only as aggregate convergence on the league norm, not as the dyadic peer copying that Mesoudi (2020) modeled directly; convergence by imitation and by a shared response to a changing environment are not fully separable in our data.

4.6 Conclusion

Treating the NFL coaching tree as a cultural pedigree, we find that the strategic traits that transmit are largely not the traits that win. Schematic identity is inherited but at most weakly selected; offensive aggression is selected but personal and, strikingly, not heritable despite being a stable individual trait; defensive aggression most clearly both transmits and wins. The negative heritability–fitness relationship of biological populations does not appear; transmission and selection are instead decoupled, and the fitness-linked behavior spreads horizontally rather than down the lineage. Coaching trees, in the end, propagate systems, not success.

Supplementary material

Table 4. Candidate feature pool for the expectation models. The game-state core is always retained; the remaining categories are subject to stability selection. Offensive models draw on up to 26 candidates (21 for two-point, which omits field position); defensive models add the two observed pre-snap indicators.

Category	Fields
Game state (core)	down, distance, yard line, score differential, quarter, seconds remaining, timeouts (both teams)
Field position	goal-to-go, side of field
Environment	home/away, division game, roof, surface, temperature, wind
Drive context	drive play count, drive first downs, net yards
Betting market	spread, total
Pre-snap (defensive models)	observed shotgun, observed no-huddle

Table 5. Within-era phylogenetic signal on the mentor network (Moran's I on a row-normalized, undirected adjacency of mentor–protege ties; one-sided permutation p from within-career-midpoint-era label shuffles). $E[I] = -1/(N-1)$ is the no-signal expectation. Under the exploratory Benjamini–Hochberg correction the defensive-aggression and tempo signals survive.

Gene	I	$E[I]$	z	p	N	edges
Offensive aggression	−0.01	−0.01	−0.04	0.52	139	440
Shotgun	+0.08	−0.01	+1.77	0.04	139	440
Tempo	+0.12	−0.01	+2.63	0.008	139	440
Defensive aggression	+0.20	−0.01	+3.03	0.002	95	216

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Competing Interests The author declares none.

Data Availability Statement The confirmatory analyses (C1–C3, hypotheses H1–H3) were preregistered before any confirmatory quantity was computed on the final measurement pipeline. All data and analysis code are public and seed-fixed, and the full pipeline runs from a single command; the preregistration, data, and code repository links are withheld here to preserve author anonymity and will be provided on acceptance.

Ethical Standards The research uses only publicly available data on professional-sport outcomes and involves no human or animal subjects; it meets all applicable ethical guidelines.

Author Contributions J.W. conceived the study, developed the methods and software, conducted the analysis, produced the visualizations, and wrote the manuscript.

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